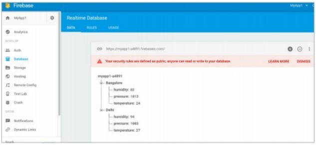

Figure 5: Checking a real-time database

the above code fragment, it writes the object's values to the Firebase *db* using *setValue()* methods that get immediately reflected in your account, as shown in Figure 5. In addition, the database rules which dictate who can read and write in real-time to the DB can be set using the *Rules* menu, as shown in Figure 5. Permissions are set as public for this illustration. You might want to create the GUI and other means of interaction with the Android app to feed in these weather values, which I'm not covering in this article as it would digress from the intended topic. I am hosting the code of a sample app at *https://github.com/shravaniv/OSFY/Firebase_sampleRTdb* for reference.

Getting notified on real-time updates

Now that data is pushed to the cloud, one can also register for real-time data updates by adding callbacks or event listeners in the app's code. The clients need to have their *google-services.json* compiled with this account's URL to get these notifications. Event listeners get notified whenever any of the JSON object's values get updated. A sample code snippet on how to add this callback for value change notifications is shown below:

```
rootRef.addValueEventListener(new ValueEventListener() {
  @Override
  public void onDataChange(DataSnapshot snapshot) {
    for (DataSnapshot postSnapshot : snapshot.getChildren())
{
        Weather climate = postSnapshot.getValue(Weather.
class);
        /* You could extract the values of object using the
getter methods
         * and display it in your GUI.

         *  climate.getCity()
         *  climate.getTemperature()
         *  climate.getHumidity()
         *  climate.getPressure()
         **/
            }
    }
    @Override
    public void onCancelled(DatabaseError firebaseError) {
       /*
        * You may print the error message.
                **/
    }
  });
```

The *onDataChange()* method gives an iterable snapshot of the database which could be used to read the updates. This way, a simple publisher-subscriber model could be implemented quickly using Firebase. There are also many other services like authentication, crash reporting, cloud messaging, analytics, etc, that are provided. You could learn and explore these different services from the Firebase site. END

References

[1] *https://firebase.google.com/docs/android/setup*
[2] *https://firebase.google.com/support/guides/firebase-android*
[3] *https://www.simplifiedcoding.net/firebase-android-tutorial-writing-firebase-data/*
[4] *https://en.wikipedia.org/wiki/Firebase*

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